

Chapter – 8 Modern Financial Theory

Before we start this chapter we need to understand basic of Option Greeks

Understanding Option Greeks

Option Greeks measure how an option's price responds to changes in market conditions.

1. Delta (Δ) – Sensitivity to Stock Price Changes

Definition: Delta measures how much the price of an option changes when the underlying stock price moves by \$1.

$$\Delta = \frac{\partial C}{\partial S}$$

- **Call Option:** Delta is **positive (0 to 1)** because the call price increases when the stock price rises.
- **Put Option:** Delta is **negative (-1 to 0)** because the put price decreases when the stock price rises.

Example:

A **call option** has a delta of **0.6**. If the stock price increases by **\$1**, the option price rises by **\$0.60**.

Example:

Hedge funds use Delta to **hedge portfolios**. If a trader owns 100 call options with a Delta of 0.5, they can hedge by short-selling 50 shares (100×0.5) to make the position Delta-neutral.

2. Gamma (Γ) – Sensitivity of Delta

Definition: Gamma measures how **Delta changes** when the stock price moves by \$1.

$$\Gamma = \frac{\partial \Delta}{\partial S}$$

- High Gamma means Delta changes **quickly**.
- Gamma is highest for **at-the-money (ATM) options** and lowest for **deep-in-the-money (ITM) or out-of-the-money (OTM) options**.

Example:

If an option's **Delta is 0.5** and **Gamma is 0.1**, a **\$1 increase** in stock price makes Delta **0.6**.

Example:

Traders in **market-making firms (e.g., Citadel, Virtu)** adjust their Delta hedges based on Gamma. If Gamma is high, frequent adjustments are needed to avoid excessive risk.

3. Theta (Θ) – Time Decay

Definition: Theta measures how much an option's price decreases per day due to **time decay**.

$$\Theta = \frac{\partial C}{\partial t}$$

- Call and put options **lose value** over time if all other factors remain constant.
- **At-the-money options** have the **highest Theta**, meaning they lose value the fastest.

Example:

A call option has **Theta = -0.05**, meaning it loses **\$0.05 per day** due to time decay.

Example:

Retail traders selling options (like **covered calls**) benefit from Theta as the option value erodes over time.

4. Vega (ν) – Sensitivity to Volatility

Definition: Vega measures how much the price of an option changes when **implied volatility** increases by 1%.

$$\nu = \frac{\partial C}{\partial \sigma}$$

- Options become more expensive when volatility increases.
- Vega is highest for **at-the-money options with a long time to expiration**.

Example:

An option has **Vega = 0.10**. If implied volatility rises by **5%**, the option price increases by **\$0.50** (0.10×5).

Example:

During the **2008 financial crisis**, option prices surged due to increased market volatility, benefiting traders who bought options before the crisis.

5. Rho (ρ) – Sensitivity to Interest Rates

Definition: Rho measures how much an option's price changes when interest rates increase by 1%.

$$\rho = \frac{\partial C}{\partial r}$$

- **Call options** increase in value when interest rates **rise** (positive Rho).
- **Put options** decrease in value when interest rates **rise** (negative Rho).

Example:

A call option has **Rho = 0.20**. If interest rates increase by **2%**, the option price increases by **\$0.40**.

Example:

In **2022**, rising interest rates affected option pricing, making call options slightly more expensive as investors discounted future cash flows differently.

Quick Summary Table

Greek	Measures Sensitivity To	Effect on Option Price
Delta (Δ)	Stock Price	Change in option price per \$1 move in stock
Gamma (Γ)	Delta	Change in Delta per \$1 move in stock
Theta (Θ)	Time	Option loses value daily due to time decay
Vega (v)	Volatility	Option price change per 1% change in volatility
Rho (ρ)	Interest Rates	Option price change per 1% change in interest rates

Modern Financial Theory

1. Black-Scholes-Merton (BSM) Model

Introduction

The Black-Scholes-Merton (BSM) model is used to price European options by assuming a risk-neutral world where assets follow a geometric Brownian motion with constant volatility.

BSM Formula

The BSM formula for a European call option price is:

The BSM formula for a European call option price is:

$$C = S_0 N(d_1) - K e^{-rt} N(d_2)$$

where:

- S_0 = Current stock price
- K = Strike price
- r = Risk-free interest rate
- t = Time to expiration
- σ = Volatility
- $N(d)$ = Cumulative standard normal distribution function

With:

$$d_1 = \frac{\ln(S_0/K) + (r + \sigma^2/2)t}{\sigma\sqrt{t}}$$

$$d_2 = d_1 - \sigma\sqrt{t}$$

Real-Life Example: Hedging at Investment Banks

Investment banks such as Goldman Sachs use the BSM model to price stock options and manage risk. Traders hedge positions using delta hedging, adjusting their portfolios continuously based on option sensitivities (Greeks).

2. Hedging and Model Completeness

Definition

- **Hedging:** Reducing risk by taking offsetting positions.
- **Model Completeness:** A market is complete if every contingent claim can be replicated by a portfolio of traded assets.

Real-Life Example: Airlines Hedging Fuel Prices

Airlines (e.g., Southwest Airlines) hedge fuel costs using derivatives. By buying call options on oil, they lock in maximum fuel costs, ensuring stability in pricing despite market fluctuations.

3. Fundamental Theorems of Asset Pricing

1. **First Theorem:** If there is no arbitrage, there exists at least one risk-neutral probability measure.
2. **Second Theorem:** If the market is complete, there is a unique risk-neutral measure.

Real-Life Example: Arbitrage in Cryptocurrency Markets

Traders exploit arbitrage opportunities by buying Bitcoin on one exchange (e.g., Binance) at a lower price and selling it on another (e.g., Coinbase) at a higher price, leading to price convergence.

4. Credit Risk Modeling

Credit risk is the risk of default by borrowers. It is quantified using models such as:

- **Standardized Approach:** Assigning fixed risk weights to assets.
- **Internal Ratings-Based (IRB) Approach:** Using bank-specific models for risk quantification.

Key Metrics

- **Probability of Default (PD):** Likelihood a borrower defaults.
- **Loss Given Default (LGD):** Percentage loss if default occurs.
- **Exposure at Default (EAD):** Amount outstanding at the time of default.

Real-Life Example: 2008 Financial Crisis

Banks underestimated PD and LGD on mortgage-backed securities, leading to massive losses when borrowers defaulted. The crisis highlighted the importance of robust credit risk models.

5. Credit Scoring Model: Z-Score Model

The **Altman Z-score** predicts bankruptcy risk:

$$Z = 1.2X_1 + 1.4X_2 + 3.3X_3 + 0.6X_4 + 1.0X_5$$

where:

- X_1 = Working capital / Total assets
- X_2 = Retained earnings / Total assets
- X_3 = EBIT / Total assets
- X_4 = Market value of equity / Total liabilities
- X_5 = Sales / Total assets

Real-Life Example: Lehman Brothers

Bankruptcy

Before Lehman collapsed, its Z-score indicated financial distress. If investors had analyzed it earlier, they might have anticipated the bankruptcy.

6. Operational Risk (OR) Modeling

Operational risk includes failures in processes, systems, or fraud.

Basel Framework

- **Basic Indicator Approach:** OR capital is a fixed percentage of gross income.
- **Standardized Approach:** Industry-specific risk multipliers are applied.
- **Advanced Measurement Approach (AMA):** Uses internal models for risk estimation.

Real-Life Example: JPMorgan's "London Whale" Trading Loss

In 2012, JPMorgan lost \$6 billion due to rogue trading by a single trader, highlighting the need for strong operational risk controls.

Let's look at some of the practical questions

1. Black-Scholes-Merton (BSM) Model

Q1: Option Pricing Calculation

A European call option has:

- Stock price (S_0) = \$50
- Strike price (K) = \$55
- Time to expiration (t) = 1 year
- Risk-free rate (r) = 5%
- Volatility (σ) = 20%

Calculate the option price using the Black-Scholes formula.

Answer

Step 2: Calculate d_1 and d_2

d_1 Calculation

$$d_1 = \frac{\ln(50/55) + (0.05 + \frac{1}{2} \times 0.20^2) \times 1}{0.20 \times \sqrt{1}}$$

$$d_1 = \frac{\ln(0.9091) + (0.05 + 0.02) \times 1}{0.20}$$

$$d_1 = \frac{-0.0953 + 0.07}{0.20}$$

$$d_1 = \frac{-0.0253}{0.20} = -0.1266$$

d_2 Calculation

$$d_2 = d_1 - \sigma\sqrt{t}$$

$$d_2 = -0.1266 - 0.20 \times 1$$

Step 3: Find $N(d_1)$ and $N(d_2)$

Using standard normal tables, we get:

$$N(d_1) = N(-0.1266) \approx 0.4496$$

$$N(d_2) = N(-0.3266) \approx 0.3723$$

Step 4: Compute the Option Price

$$C = 50 \times 0.4496 - 55 \times e^{-0.05 \times 1} \times 0.3723$$

First Term:

$$50 \times 0.4496 = 22.48$$

Second Term:

$$e^{-0.05} = 0.9512$$

$$55 \times 0.9512 = 52.32$$

$$52.32 \times 0.3723 = 19.47$$

$$C = 22.48 - 19.47 = 3.01$$

Q2: Hedging with Delta

An investor holds a portfolio containing 1,000 European call options on a stock. If the delta of each option is 0.6, how many shares should the investor hold to create a delta-neutral portfolio?

Answer

$$\begin{aligned}\text{Hedging Portfolio} &= -\text{Delta} \times \text{Number of Options} \\ &= -0.6 \times 1000 = -600\end{aligned}$$

The investor should short **600 shares** to be delta-neutral.

Q3: Implied Volatility Concept

If two options have the same strike price and time to expiration, but one is more expensive than the other, what can you infer about their implied volatilities?

Answer

The more expensive option has **higher implied volatility**, as a higher volatility increases the probability of the option finishing in the money.

2. Credit Risk Modeling

Q4: Probability of Default (PD) Calculation

A bank estimates that the **PD** of a corporate borrower is **2%** per year. If the **EAD** is **\$5 million** and **LGD** is **40%**, calculate the expected credit loss (ECL).

Answer

$$\begin{aligned} ECL &= PD \times LGD \times EAD \\ &= 0.02 \times 0.40 \times 5,000,000 \\ &= 40,000 \end{aligned}$$

The expected credit loss is \$40,000.

Q5: Credit Scoring using Z-Score

A company has the following financials:

- Working Capital / Total Assets = **0.2**
- Retained Earnings / Total Assets = **0.1**
- EBIT / Total Assets = **0.15**
- Market Value of Equity / Total Liabilities = **0.5**
- Sales / Total Assets = **0.7**

Using Altman’s Z-score formula:

$$Z = 1.2X_1 + 1.4X_2 + 3.3X_3 + 0.6X_4 + 1.0X_5$$

Find the company’s Z-score.

Answer

$$\begin{aligned} Z &= 1.2(0.2) + 1.4(0.1) + 3.3(0.15) + 0.6(0.5) + 1.0(0.7) \\ &= 0.24 + 0.14 + 0.495 + 0.3 + 0.7 \\ &= 1.875 \end{aligned}$$

Since $Z < 2.99$, the company is in financial distress and may have a high probability of bankruptcy.

Calculation as per Excel:

Variable	Value	Weight	Weighted Value
X1 (Working Capital / Total Assets)	0.2	1.2	0.24
X2 (Retained Earnings / Total Assets)	0.1	1.4	0.14
X3 (EBIT / Total Assets)	0.15	3.3	0.495
X4 (Market Value of Equity / Total Liabilities)	0.5	0.6	0.3
X5 (Sales / Total Assets)	0.7	1	0.7
Z-Score			1.875

Interpretation of the Z-Score

$Z > 2.99 \rightarrow$ **Safe Zone** (Low risk of bankruptcy) ✓

$1.81 \leq Z \leq 2.99 \rightarrow$ **Gray Zone** (Medium risk, could go either way) ⚠

$Z < 1.81 \rightarrow$ **Distress Zone** (High probability of bankruptcy) ✗

Since $Z = 1.875$, the company falls into the **Gray Zone**, meaning:

The company is **not safe**, but it is **not immediately at risk** either.

It should **improve financial health** to avoid slipping into the **Distress Zone** (< 1.81).

Focus on **increasing retained earnings, profitability (EBIT), and working capital** to improve the score.

Q6: IRB vs. Standardized Approach

A bank is evaluating two credit risk assessment approaches:

1. **Standardized Approach** assigns a fixed **risk weight** of **75%** to retail loans.
2. **Internal Ratings-Based (IRB) Approach** estimates **PD = 1%**, **LGD = 50%**, and **EAD = \$2 million**.

Which approach results in lower risk-weighted assets (RWA) for the loan?

Answer

- Standardized Approach

$$RWA = 75\% \times 2,000,000 = 1,500,000$$

- IRB Approach

$$\begin{aligned} RWA &= PD \times LGD \times EAD \\ &= 0.01 \times 0.50 \times 2,000,000 = 10,000 \end{aligned}$$

Since IRB RWA (10,000) < Standardized RWA (1,500,000), the bank benefits from using IRB.